

Homework: Blockchain Tech

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TOPIC: SECP256K1 IN BLOCKCHAIN CRYPTOGRAPHY

Abstract

Elliptic Curve Cryptography (ECC) is widely used to secure blockchain systems. SECP256k1 is an elliptic curve used in Bitcoin and many other blockchains for key generation and digital signatures. This paper briefly explains what SECP256k1 is, how it works mathematically, why it was chosen for Bitcoin, and its security and efficiency. A related research study on elliptic curve cryptography is also reviewed to show its importance in blockchain security.

Keywords—SECP256k1, Blockchain, Elliptic Curve Cryptography, Digital Signature

I. Introduction

Blockchain technology depends on cryptography to protect transactions and user identities. Instead of using a central authority, blockchain systems rely on mathematical security. Elliptic Curve Cryptography is commonly used because it provides strong security with smaller keys. SECP256k1 is one of the most important elliptic curves used in blockchain systems, especially Bitcoin.

II. Overview of SECP256k1

SECP256k1 is an elliptic curve defined over a finite field and mainly used with the Elliptic Curve Digital Signature Algorithm (ECDSA). It allows users to generate a private key and a public key. The private key is kept secret, while the public key is shared to verify transactions. This mechanism ensures that only the real owner can authorize a blockchain transaction [1].

III. Mathematical Foundation

SECP256k1 is based on elliptic curve mathematics, using equations of the form:

$$y^2 = x^3 + ax + b$$

Its security depends on the Elliptic Curve Discrete Logarithm Problem, which is very difficult to solve. A private key is randomly generated, and a public key is created by multiplying a point on the curve. Digital signatures are produced using the private key and verified using the public key, ensuring message integrity and authenticity.

IV. Why SECP256k1 Is Used in Bitcoin

Bitcoin uses SECP256k1 because it is efficient and secure. It allows faster signature generation and verification compared to some other curves. In addition, its parameters are clearly defined and not randomly generated, which increases trust in its security. These features make SECP256k1 suitable for decentralized blockchain systems.

V. Security and Efficiency

SECP256k1 provides strong security while using smaller key sizes, which improves performance. It supports important security features such as authentication, data integrity, and non-repudiation. Although quantum computing may pose future risks, SECP256k1 is currently considered secure and reliable for blockchain use.

VI. Related Research

The [SEC 2: RECOMMENDED ELLIPTIC CURVE DOMAIN PARAMETERS](#) document by the Standards for Efficient Cryptography Group explains the technical design and security of elliptic curves such as SECP256k1. This research shows why ECC is effective for digital signatures and secure communication, making it essential for blockchain security [2].

VII. Conclusion

SECP256k1 is a key cryptographic component in blockchain systems. Its strong mathematical foundation, efficiency, and security make it ideal for digital signatures and transaction

verification. As a result, SECP256k1 plays an important role in maintaining trust and security in blockchain networks.

References

[1] S. Sek, [CRYPTOGRAPHY FOR BLOCKCHAIN](#), Faculty of Digital Economy, National University of Management, Cambodia, 2025.

[2] Standards for Efficient Cryptography Group (SECG), [SEC 2: RECOMMENDED ELLIPTIC CURVE DOMAIN PARAMETERS](#), Version 2.0, 2010.